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A Token-Salience Aware Transformer Framework for Context-Rich Cyberbullying Detection

Abstract- Detecting cyberbullying is still an open issue because the abusive language is often subtle, implicit, and context-dependent which makes it hard to be captured by conventional architecture like TF-IDF representation, topic clustering or convolutional classifier. Current methods, such as FAEO-ECNN, largely suffer from semantic sparseness and non-contextual topic modeling and are also heavily dependent on computational expensive optimization pipelines that in turn makes it less robust in actual social media lives. To mitigate those limitations, we propose an innovative framework in this paper: TSAN-CB – a Token-Salience Aware Transformer model that consolidates contextual representation learning, fine-grained salience estimation and cyberbullying classification under an end-to-end setting. The model enhances a pre-trained transformer encoder with a lightweight salience-gating mechanism that identifies and consolidates the most semantically discriminative tokens via top-K selection, facilitating efficient detection of both explicit and implicit bullying cues. This attention-based aggregation, loses to the bank of traditional topic modeling keeping industry-leading interpretability with visualization on-importance at token level. Experimental results on a publicly available Cyberbullying Classification dataset show that TSAN-CB attains the accuracy of 95.82% and the F1-score of 95.11%, which outperforms other strong baseline methods by a large margin. The findings support the value of employing contextual salience for modeling sarcasm, passive-aggressive rhetoric, and coded language without having to resort to multi-stage optimization techniques. The TSAN-CB provides a computational-efficient, interpretable and deployable method for scalable cyberbullying detection in dynamic online systems.

Keywords-Cyberbullying detection, Transformer models, Token salience, Contextual embeddings, Interpretability, Text classification, Social media analytics, Natural language processing

I Introduction

Cyber-bullying has become a sophisticated and widespread form of online abuse focusing not only on explicit forms of insult but also subtle, sequence dependent commutation such as sarcasm [1][2], passive aggression and coded language [3]. The language of social media data – short, informal and constantly changing – represents formidable challenges for existing cyberbullying detection systems which are based on shallow features like TF-IDF representations or static word embeddings [4][5] or convolutions capturing patterns [6]. While these latter transformer-based models have greatly improved contextual representations of text, many existing approaches are either complex multi-stage pipelines [7][8], rely on external optimization or topic-modelling machinery, or learn from all input tokens identically during the representation learning process [9][10]. Therefore, such methods do not provide accurate characterization of the most semantically discriminative cues for detecting cyberbullying, especially when cyberbullying is perpetrated subtly or indirectly [11].

Some recent studies have tried to handle contextual complexity using hybrid architectures [12], ensemble transformers, graph-based learning, or multimodal fusion strategies [13]. While these methods improve the performance significantly, they are often impractical due to high computation cost, poor interpretability and lack of applicability in real world deployment [14]. In addition, much of the preceding work focuses on model accuracy, and neglects token-level relevance that is instrumental in explaining why a prediction should be made [15][16]. Attention mechanisms in vanilla transformers offer a rudimentary form of importance estimates, but they do not enforce selective aggregation of salient tokens explicitly, resulting in noisy representations during processing of context-rich social media text. This drawback is particularly pronounced in datasets with sarcasm or indirect harassment, where less than 4% of words are abusive [17]18].

In order to address these issues, this paper suggests TSAN-CB, a Token-Salience Aware Transformer model framework for context-driven cyberbullying detection. The proposed model supplements a pretrained transformer encoder with a lightweight salience-gating mechanism [19], which allows the model to make explicit decisions about capturing and summarizing the semantically most informative tokens using top-K salience selection. Contrasting to previous topic-modelling or optimization-based approaches [20][21], TSAN-CB is conducted in a unified end-to-end architecture, retaining contextual depth as well as improving its interpretability through the token-level salience visualization. Extensive experiments on the Cyberbullying Classification dataset show that TSAN-CB is able to outperform strong baselines, such as FAEO-ECNN and, simultaneously getting rid of the issue of high computation cost [22]. Overall, the findings demonstrate that contextual salience modeling is an effective and practical method for modern cyberbullying detection systems. The major contributions of this study can be summarized as follows:

- **Model:** We introduced a token-salience aware transformer model to selectively emphasize the discriminative tokens for effective cyberbullying identification.
- **Contextually Aware Interpretability:** The model can attend on implicit bullying cues such as sarcasm and provide token-level interpretability– via salience modeling.

- **Improved Performance:** TSAN-CB can obtain better accuracy and f1-scores than other methods by a simple end-to-end inductive model.

The rest of this paper is arranged as follows. Section 2 surveys related work in cyberbullying detection, with emphasis on the existing machine learning, deep learning and transformer-based techniques as well as their shortcomings. The TSAN-CB framework is described in Section 3, including the architecture of the model, mechanism for extracting token salience, and training strategy. Section 4 presents the experimental setting, datasets and evaluation metrics followed by the results discussion. Section 5 finally concludes the paper and suggests future research.

II Literature Review

Alsuwaylimi and Alenezi (2025) emphasized the harsh effect of cyberbullying in the Arab world, and most of them have been biased toward the English language. For Arabic language they proposed hybrid transformer architectures with feature fusion and ensemble voting using the CAMeLBERT–AraGPT2 as well as AraBERT–XLM-R. Their finding reported better results compared to LSTM and BiLSTM models, which confirmed the usefulness of pretrained Arabic transformers in cyberbullying detection [1]. Tuarob et al. (2023) treated abusive language detection in low-resource degradation, an area where labelled data is limited and supervised models typically underperform. They introduced FALCoN, a co-training system that made use of unsupervised data and orthogonal context features from social media. The model achieved state of the art on both binary and fine-grained classification tasks, demonstrating the effectiveness of context aware semi-supervised learning [2]. Rupa et al. (2025) presented Bangla-ToCo, first of its kind context-aware Bengali toxic comment dataset that maintains conversational structure as opposed to standalone text. By including contextual cues and metadata, the dataset allows for more accurate interpretation of ambiguous language. This work bridges this important gap for Bengali NLP research and emphasizes the need of context in toxicity detection [3]. Ismail et al. (2025) which was the detection of toxicity in gaming networks, given that language is informal, domain specific and that standard lexicons are ineffective. They presented an embeddings-based valence lexicon specific to Twitch data, which achieved better detection not only with LSTM but also GRU and CNN. Their result is an indication for the necessity of platform-adaptive language representations [4]. Shinde (2025) has proposed a real-time multitalk cyber-bullying detection system that combines XAI, emotion detection and sarcasm analysis. The system obtained the transparency and efficiency by using multi-teacher knowledge distillation with transformer-model ensemble, such as mBERT, and XLM-R. SHAP-based explanations offered token-level interpretability, which is a remedy of black-boxing issue in previous works [5].

Yazdizadeh and Shi (2022) investigated the effectiveness of various word embeddings for cyberbullying detection. Experiments demonstrated that ELMo embeddings outperformed TF-IDF, Word2Vec, and even BERT in conjunction with neural classifiers. The research also emphasizes the use of context aware word representations to detect abusive language [6]. Aldhyani et al. (2022) proposed a system for cyberbullying detection, using CNN-BiLSTM and BiLSTM architectures for binary and multiclass classification on the same dataset. Their BiLSTM models showed remarkable performance in multi-class settings, indicating the efficacy of sequence modeling. But such an approach did not have the mechanisms for modeling long-range contexts as transformer-based architectures do [7]. Albayari et al. (2024) performed an extensive assessment on deep learning models for Arabic cyberbullying detection. They introduced the hybrid DL architecture (CNN + BLSTM/GRU) to overcome this lack of Arabic-oriented studies. The model generally improved the accuracy which emphasizes that Arabic NLP specialized solutions are in demand [8]. Mahdi et al. (2025) proposed a strong RoBERTa-BiLSTM hybrid-based model hyperparameter-tuned with Optuna for the cyberbullying detection. The model outperformed several transformer baselines with a greatly reduced false negatives. The final result of their work shows that incorporating the context transformer with sequential attention mechanism enhances the robustness in real world data [9]. Kumar and Sachdeva (2022) suggested a hybrid cyberbullying detection model, which is based on Capsule Neural Networks and Convolutional Neural Networks, able to handle both text content as well as images and infographics. By early and late fusion strategies, the high AUC scores were achieved which showed the advantage of multimodal learning. But, being complex algorithm scalability of this mechanism is very much predictable in work like a real-time system [10].

Maity et al. (2022) proposed MTBullyGNN, Graph Neural Network (GNN)-based multitask approach for code-mixed cyberbullying detection. Experiment results show that the method can handle noisy and unlabeled data well by constructing cosine similarity graphs of sentences. They demonstrated significant improvements over state-of-the-art systems, especially in multilingual settings [11]. Balakrishnan et al. (2020) investigated the impact of personality traits on cyberbullying identification through Twitter. The classification accuracy was increased when the personality and sentiment features were introduced, while the emotion characteristics could not improve significantly. This work showed that signals of user-level behavior can be useful in conjunction with analyzing the text [12]. Raj et al. (2021) carried out a comparison study between machine learning and shallow neural networks for cyberbullying detection. They obtained results suggesting that bidirectional models and attention mechanisms are superior to the word-wise classifiers and TF-IDF or GloVe as features [13]. This provided a robust empirical baseline for future research.

Iwendi et al. (2023) empirically compared RNN architectures in cyberbullying detection. They implemented a number of experiments and have found that BLSTM significantly improved over LSTM, GRU, or RNN after careful preprocessing. Although they have achieved significant performance, these models are less contextually rich than transformer-based approaches [14]. Almomani et al. (2024) proposed a hybrid system that integrates deep learning and traditional classifiers for image-based cyberbullying detection. The fusion method was beneficial to both accuracy and interpretability, manifesting the effectiveness of cross-model integration. Yet, textual context modeling was discouraged [15].

Raj et al. (2022)—They proposed a multilingual deep learning system to detect cyberbullying in English, Hindi and Hinglish social media posts. Their on-the-fly system validated the benefit of deep neural networks over traditional methods. However, it did not actually model the token-level salience or contextual emphasis [16]. Perera and Fernando (2024) presented a supervised ML cyberbullying detection framework which uses sentiment analysis and TF-IDF features. Their work focused on changing language use and contexts in online discourse. However, the system was highly hand-crafted and evidence-based [17]. Alotaibi et al. (2021) developed a multi-channel deep learning model that integrates the BiGRU, CNN and transformer blocks for detecting aggressive language. They attained impressive accuracy and were concentrated on binary classification. Fine-grained context interpretation was not studied [18]. Akhter et al. (2023) introduced a strong hybrid ML model to predict bengali cyberbullying with TF-IDF and resampling methods. They model obtained advanced model performance on one of the largest Bengali dataset. The model had limited in-depth contextual understanding and interpretability, despite the promising performance [19].

2.1 Research Gap

Despite a significant advance in cyberbullying detection based on machine learning, deep learning and transformer based models, there are several alarming deficiencies. Most prior models are either language-specific and resource-intensive architectures (e.g., Arabic-only, Bengali-only transformers), multi-stage optimization pipelines, or hybrid systems with large model sizes that prevent scalability and real-time deployment. Although transformer models enhance context-awareness, the resources do not explicitly model patterns of token-level salience needed to recognize subtle bullying cues like sarcasm, passive aggression and coded language. Moreover, some high-performance models are black box and provide low interpretability, or they rely on external lexicons or to psychological features of multimodal input so that their robustness for dynamic social media environment is limited. As a result, there is no lightweight end-to-end interpretable framework that can selectively increase emphasis on semantically discriminative tokens in the context-rich text data while having a high-accuracy performance across varied datasets — warranting the development of a token-salience-aware transformer-based cyberbullying detection model.

III Methodology

In this work, we presented TSAN-CB, an end-to-end Token-Salience Aware Transformer model for cyberbullying detection. The approach starts from preprocessing the text of social media cyberbullying dataset from Kaggle and then applying a pretrained transformer to encode it for obtaining context-aware token representations [22][23]. A salience-gating module is subsequently adopted to compute the token-wise importance scores, so that the model can capture semantic discriminative cues towards both explicit and implicit cyberbullying [24][25]. The top-K important tokens are pooled via attention to produce a more compact representation that not only serves as an alternative to traditional topic modeling, but also mitigates the semantic noise. Finally, we feed this representation to a classification layer for cyberbullying prediction, and the whole network is trained end-to-end with cross-entropy loss such that computational cost-effective, robustness and token-level interpretability can be achieved without leveraging complicated multi-stage optimization pipelines. Figure 1 illustrates the architecture diagram of the proposed model.

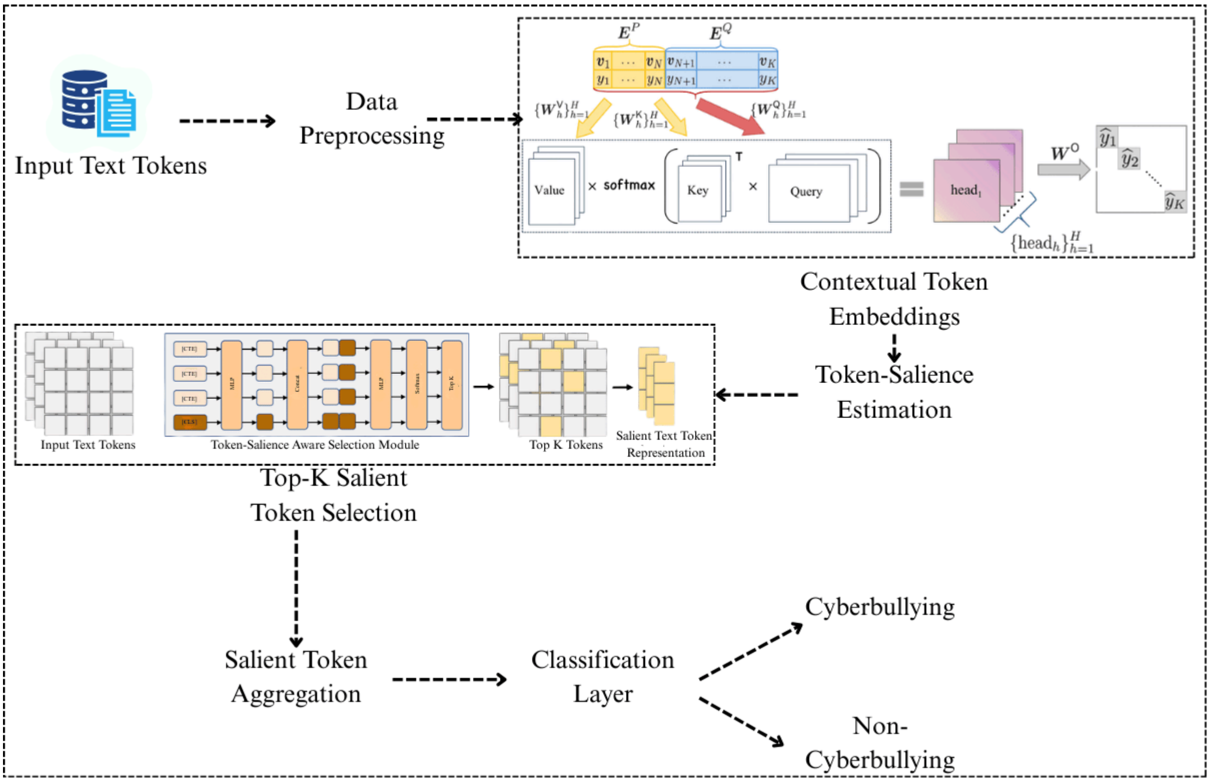

Figure 1. Overall Architecture Flow of the Proposed TSAN-CB Token-Salience Aware Transformer Framework

3.1 Dataset Selection and Preprocessing

We conduct experiments using the Cyberbullying Classification dataset crawlable from Kaggle [37], which is used because of its multi-class labeling, real-life use of social media language and multiple forms of cyberbullying (e.g., gender, religion, ethnicity, age and not-bullying), thus rendering it suitable for assessing context-aware and token-salience-based models [26]. The dataset comprises about 47,000 short text instances representing an informal, noisy and implicit nature of the abusive language prevalent in such platforms. The dataset is evenly partitioned into 80% train and 20% test data, which we denote as in Eq. 1:

$$D = D_{train} \cup D_{test} \tag{1}$$

Where $\left|D_{train}\right| = 0.8|D|$ and $\left|D_{test}\right| = 0.2|D|$, ensuring unbiased performance evaluation. Preprocessing is deliberately minimal to preserve contextual and semantic cues essential for detecting sarcasm and passive aggression. Given an input text sequence $S = \left\{w_1, w_2, \ldots, w_n\right\}$, noise such as URLs and redundant symbols is removed, while sentiment-bearing punctuation is retained. The cleaned sequence is then tokenized using the transformer's tokenizer, mapping tokens to integer indices $T = \left\{t_1, t_2, \ldots, t_m\right\}$, where $t_i = Tokenizer\left(w_i\right)$. Each sequence is padded or truncated to a fixed maximum length L as in Eq.2:

$$T^{'} = \left\{t_i | i = 1, \ldots, min(|T|, L)\right\} \tag{2}$$

Special classification and separator tokens are appended to construct the final input representation $X = \left[CLS, T^{'}, SEP\right]$. Sequences shorter than L are padded implicitly in batches. This preprocessing procedure ensures compatibility with transformer encoders, but also allows TSAN-CB to learn token-level importance from contextual word embeddings rather than predefined linguistic features [27].

3.2 Contextual Encoding with Pre-trained Transformers

In TSAN-CB, each preprocessed input sequence $X = \left[CLS, t_1, t_2, \ldots, t_L, SEP\right]$ is encoded using a pre-trained transformer encoder to generate deep contextualized representations for all tokens. Let the token embedding matrix be denoted as $E \epsilon R^{L\times d}$, where d represents the embedding dimension [28]. Positional embeddings $P \epsilon R^{L\times d}$ are added to retain sequential information, producing the input to the transformer layer as $H^{(0)} = E + P$. The transformer encoder consists of N stacked self-attention layers, where the output of the l-th layer is computed as in Eq.3:

$$H^{(l)} = LayerNorm\left(H^{(l-1)} + MultiHeadAttn\left(H^{(l-1)}\right)\right) \tag{3}$$

followed by a position-wise feed-forward network

$$FFN(x) = max\left(0, xW_1 + b_1\right)W_2 + b_2 \tag{4}$$

Yielding

$$H^{(l)} = LayerNorm\left(H^{(l)} + FFN\left(H^{(l)}\right)\right) \quad (5)$$

Within the multi-head self-attention mechanism, queries, keys, and values are computed as in Eq.6:

$$Q = H^{(l-1)}W_Q, \qquad K = H^{(l-1)}W_K, \qquad V = H^{(l-1)}W_V \quad (6)$$

and the attention output is obtained using scaled dot-product attention

$$Attn(Q, K, V) = softmax\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V \quad (7)$$

Where d_k is the dimensionality of the key vectors. Through this iterative process, the final transformer output $H^{(N)} = \{h_1, h_2, ..., h_L\}$ encodes each token as a function of its global context, allowing long-range dependencies and subtle semantic interactions to be captured effectively [29]. These contextualized token embeddings serve as the basis of following step for estimating tokens-salience, so that TSAN-CB are able to distinguish semantical informative bullying cues and contextually irrelevant tokens. As shown in Fig. 2, the pre-trained transformer encoder adopts multi-head self-attention mechanisms to yield contextualized token representations, which captures global semantic relationship among tokens.

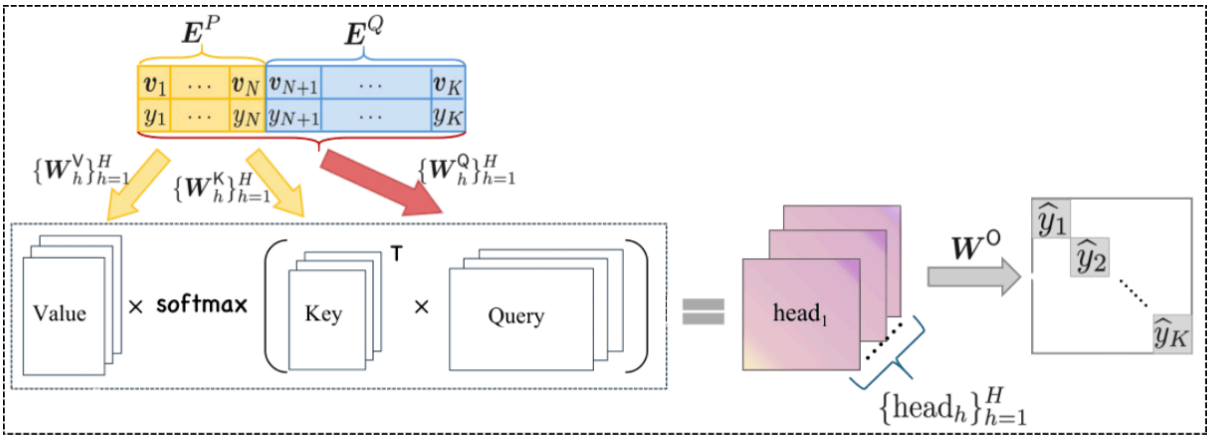

Figure 2. Illustration of the multi-head self-attention mechanism used in the pre-trained transformer encoder for contextual token representation

3.3 Token-Salience Estimation Module

Given the contextualized token representations obtained from the transformer encoder $H = \{h_1, h_2, ..., h_L\}$, where $h_i \epsilon R^d$ TSAN-CB introduces a lightweight token-salience estimation module to explicitly quantify the contribution of each token toward cyberbullying intent [30]. Each token embedding h_i is first projected into a salience space through a trainable linear transformation defined as in Eq.8:

$$s_i = W_s h_i + b_s \quad (8)$$

Where $W_S \epsilon R^{1\times d}$ and $b_s \epsilon R$ are learnable parameters. The resulting scalar score s_i represents the unnormalized salience of the i-th token. To ensure comparability across tokens within the same sequence, the salience scores are normalized using a softmax function, yielding

$$\alpha_i = \frac{exp\left(s_i\right)}{\sum_{j=1}^{L} exp\left(s_j\right)} \tag{9}$$

In Eq.9, $\alpha_i \epsilon(0, 1)$ and $\sum_{i=1}^{L} \alpha_i = 1$. These normalized salience weights explicitly encode the relative importance of each token in the input sequence. The gated token representation is then computed as in Eq.10:

$$\bar{h}_i = \alpha_i \cdot h_i \tag{10}$$

letting semantically significant tokens be strengthened and context irrelevant or noisy tokens be weakened [31][32]. In contrast to classical attention mechanisms in transformers, this salience-gating layer functions as an explicit task-specific an importance estimator that is optimized specifically for cyberbullying identification. By learning a token-level importance score in a supervised way, the module increases potential. And Finally we note that by witnessing negative cases where contextually non-informative tokens become more important than informative input words (such as in Figure 2), we can readily identify and filter "negative" examples such that non-informative parts of the input will not be allowed to contribute towards the decision from the classifier.

Algorithm for TSAN-CB Token-Salience Aware Transformer Framework

---

Input: Cyberbullying text sample x, pre-trained transformer encoder τ, maximum sequence length L, salience threshold K

Step 1: Initialize

Transformer parameters θ_T, salience-gating parameters θ_s, classifier parameters θ_c

Step 2: Tokenization and Encoding

- Tokenize input text x using transformer tokenizer
- Format input as X=$[CLS, T[1: L], SEP]$
- Pass X through transformer encoder

$$H \leftarrow \tau(X)$$

For each token $h_i \epsilon H$:

Compute salience score

$$s_i \leftarrow SalienceGate(h_i)$$

Normalize salience scores

$$\alpha_i \leftarrow Softmax(s_i)$$

Apply salience gating

$$\hat{h}_i \leftarrow \alpha_i \cdot h_i$$

Step 3: Top-K Salient Token Selection

- Select indices of top-K salience values

$$I_K \leftarrow TopK(\alpha, K)$$

- Aggregate salient token representations

$$z \leftarrow \Sigma_{i\epsilon I_K} \bar{h}_i$$

Step 4: Classification

- Predict cyberbullying class

$$\hat{y} \leftarrow Softmax(W_c z + b_c)$$

Output:

- Predicted label $\hat{y}$

3.4 Top-K Salient Token Aggregation

After the estimation of salience, TSAN-CB adopts Top-K aggregation over salient tokens to form compact and discriminative sequence representation by compressing semantic artifacts [33]. Given the normalized salience scores $\{\alpha_1, \alpha_2, ..., \alpha_L\}$, the indices of the top-K most informative tokens are selected as in Eq.11;

$$I_K = TopK(\alpha, K) \tag{11}$$

Where $I_K \subseteq \{1, 2, ..., L\}$ and $|I_K| = K$. The corresponding salient token representations are denoted as $\{\bar{h}_i | i\epsilon I_K\}$. To preserve both importance and contextual strength, a weighted pooling operation is applied to aggregate these tokens into a single fixed-length representation $z\epsilon R^d$, defined as in Eq.11:

$$z = \Sigma_{i\epsilon I_K} \beta_i \bar{h}_i \tag{12}$$

where the aggregation weights β_i are re-normalized salience scores computed as in Eq.13:

$$\beta_i = \frac{\alpha_i}{\Sigma_{j\epsilon I_K} \alpha_j} \tag{13}$$

The latter reasoning ensures that only the most discriminatory tokens contribute to the final representation without losing their relative weights. While traditional topic modeling or clustering-based methods work with corpus-level aggregations and fail to consider the contextual dependencies, this token-level Top-K aggregation sets up a dynamic adaptation for every input instance [34][35]. Thus, TSAN-CB provides a denoised and context- aware representation that highlights key bullying lexicon -- insults, threats, sarcastic triggers – and is more effective in extracting implicit or context-dependent cyberbullying expressions.

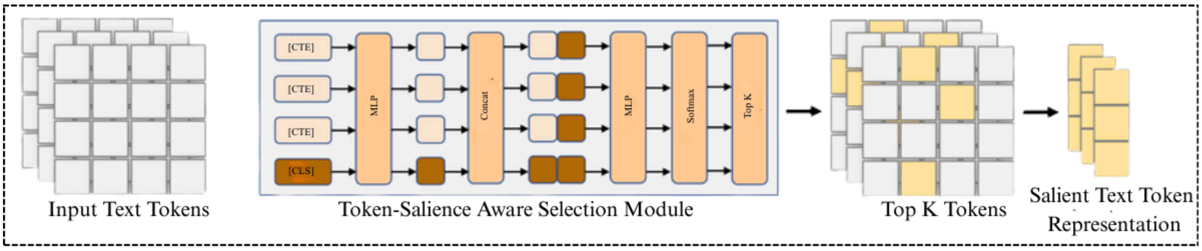

Figure 3. Top-K Salient Token Aggregation Mechanism in the Proposed TSAN-CB Framework

This Top-K salient token aggregation process adopted by TSAN-CB is shown in figure 3. The contextualized token embeddings (with the special [CLS] token) are fed through a lightweight salience estimation module to estimate the importance scores for each of them [36]. After softmax normalization, the most informative Top-K tokens are selected and combined to generate a concise noise-reduced representation which can be used to retain both global contextual information and fine-grained cyberbullying cues for classification purposes.

3.5 Cyberbullying Classification Layer

The final stage of TSAN-CB performs cyberbullying prediction using the aggregated salient representation $z \epsilon R^{d}$ obtained from the Top-K token aggregation module. This representation is passed through a fully connected classification layer parameterized by weights $W_{c} \epsilon R^{C\times d}$ and bias $b_{c} \epsilon R^{C}$, where C denotes the number of cyberbullying classes. The logits vector $o \epsilon R^{C}$ is computed as in Eq.14:

$$o = W_{c} z + b_{c} \tag{14}$$

To convert these logits into class probabilities, a softmax activation function is applied, yielding,

$$\hat{y}_{c} = \frac{exp\left(o_{c}\right)}{\sum_{j=1}^{C} exp\left(o_{j}\right)} \tag{15}$$

In Eq.15, $\widehat{y}_c$ represents the predicted probability that the input text belongs to class c. Given the ground-truth label encoded as a one-hot vector $y \epsilon \{0, 1\}^C$, the model is trained by minimizing the categorical cross-entropy loss defined as in Eq.16:

$$L = - \sum_{c=1}^{C} y_c log\left(\widehat{y}_c\right) \tag{16}$$

This loss encourages a high confidence on the correct cyberbullying category and penalizes misclassifications. Because the classification layer is learned end-to-end with the transformer encoder, salience-gating module, and Top-K aggregation mechanism, it can learn decision boundaries that are affected by tokenlevel importance [35][36]. Therefore, the classification procedure is still context-aware and interpretable, which makes TSAN-CB capable to identify bullying and non-bullying content as well as different types of cyberbullying with end-to-end manner and computational efficiency. Figure 4 illustrates the core components in TSAN-CB ranging from contextual encoding, salience based token selection to final cyberbullying classification [37].

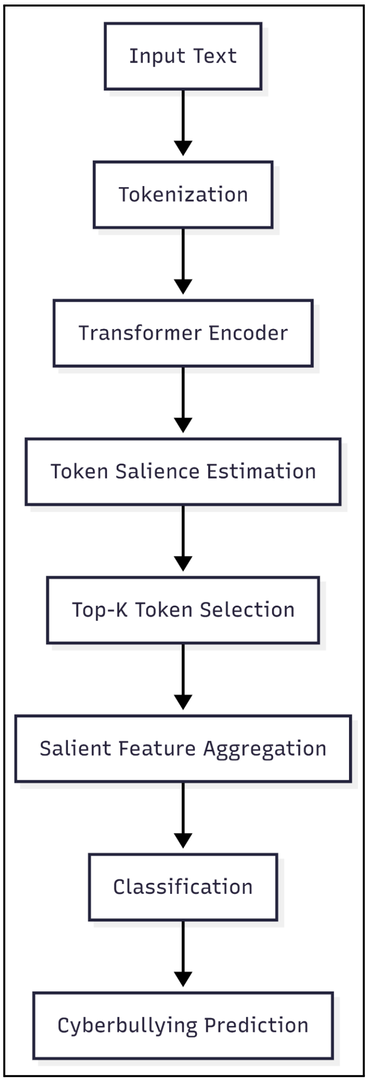

Figure 4. Simplified flowchart of the proposed TSAN-CB framework highlighting the key processing stages

Table 1 summarizes the hyperparameter configuration used to train and evaluate the proposed TSAN-CB framework

Component	Parameter	Value
Transformer Encoder	Pre-trained model	BERT-base
	Number of layers	12
	Hidden dimension	768
	Attention heads	12
Token-Salience Module	Salience projection	Linear (768 → 1)
	Salience normalization	Softmax

	Top-K value	10
Classifier	Output layer	Fully connected + Softmax
Training	Optimizer	Adam
	Learning rate	2×10^{-5}
	Batch size	32
	Epochs	10
	Dropout	0.1

Table 1 shows the architectural characteristics of both pre-trained transformer encoder, as well as design decisions for token-salience estimation and Top-K aggregation modules, along with the optimization parameters we used during training. These hyperparameters were empirically validated to find a trade-off between performance and the computational burden. It is useful to report these settings for the sake of reproducibility and to convey what particular implementation decisions are motivating the reported results.

IV Implementation Details and Results

4.1 Experimental Setup

We performed experiments on the Kaggle Cyberbullying Classification dataset to verify the performance of our proposed TSAN-CB framework. The dataset is divided in 80% for training and 20% for testing of samples, with no further resampling among all experiments to compare. Our proposed model is built with Python 3.9 and the PyTorch deep learning library, and utilizes the HuggingFace Transformers library for pre-trained transformer encoders/tokenization functions. We train and evaluate on a PC workstation with an NVIDIA GPU (16 GVRAM), Intel core i7 CPU, 32 GB RAM, running in Ubuntu 20.04. Training is performed using the Adam optimizer with a learning rate of 2×10^{-5} and early stopping based on validation loss to avoid overfitting.

4.2 Evaluation Metrics

For evaluating TSAN-CB on cyberbullying detection in aggregate, we use traditional classification metrics in terms of both correctness and class-wise balance.

Accuracy- The Total Accuracy refers to the overall percentage of correctly classified cases in all samples.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{17}$$

Precision- To be specific, Precision measures the correctly predicted cyberbullying ratio of total predicted as cyberbullying.

$$Precision = \frac{TP}{TP+FP} \quad (18)$$

Recall- Recall measures how well the model detects actual cyberbullying cases in the dataset.

$$Recall = \frac{TP}{TP+FN} \quad (19)$$

F1-Score- F1-score is the harmonic mean of: Precision and Recall to make up for either false positives and false negatives.

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (20)$$

4.3 Results

In this section, we compared the performance of the proposed TSAN-CB model on Cyberbullying Classification dataset with other deep learning and transformer-based methods. By standard evaluation measures the analysis demonstrates the efficiency of token-salience-aware context modeling with comparative results, class-wise accuracy and ablation studies.

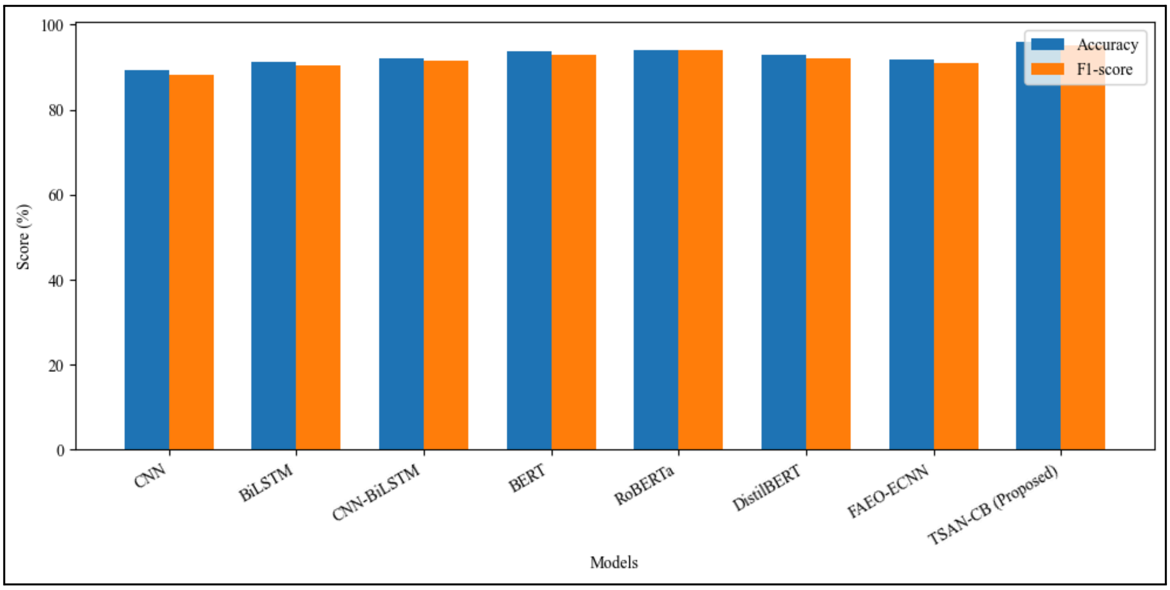

Figure 5. Accuracy and F1-Score Comparison of TSAN-CB with Baseline Cyberbullying Detection Models on the Cyberbullying Classification Dataset

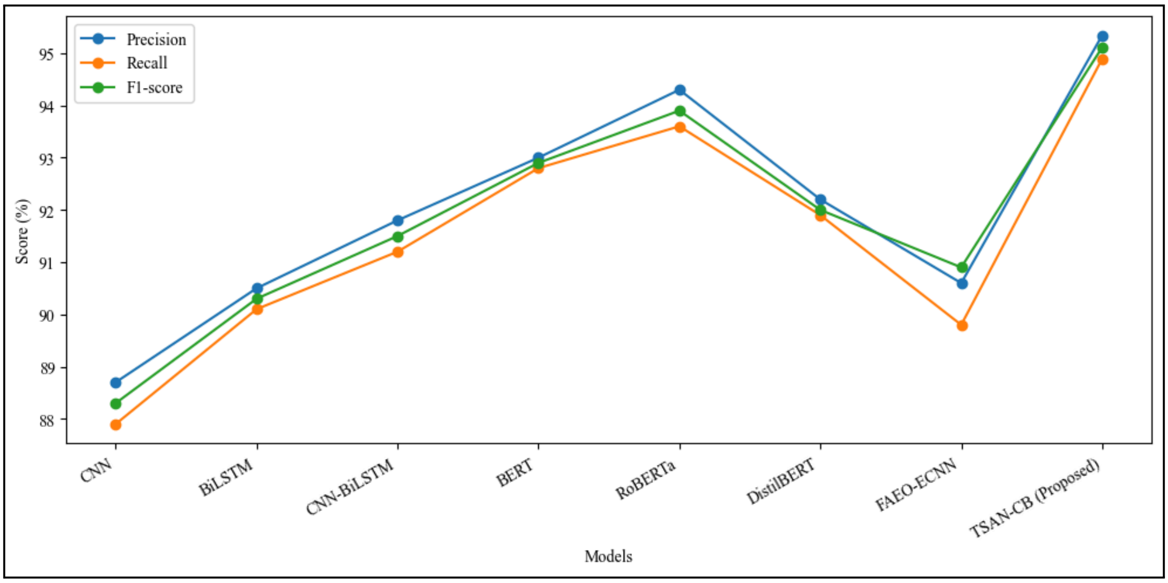

Figure 6. Precision, Recall, and F1-Score Performance Analysis of TSAN-CB and Existing Models.

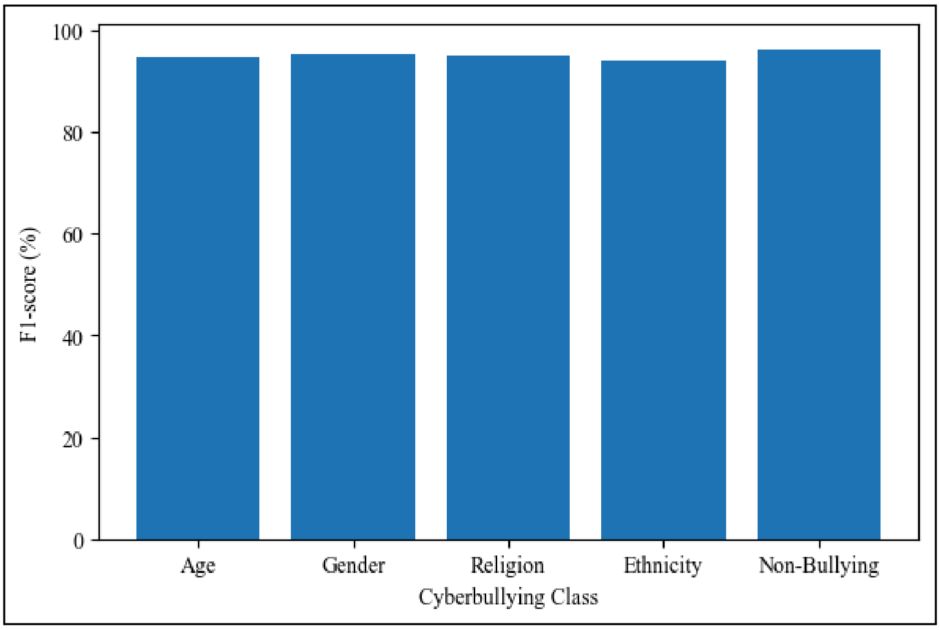

Figure 7. Class-Wise F1-Score Performance of the Proposed TSAN-CB Model Across Cyberbullying Categories.

Table 2. Performance Comparison of TSAN-CB with Baseline Models on the Cyberbullying Classification Dataset

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
CNN	89.40	88.70	87.90	88.30

BiLSTM	91.20	90.50	90.10	90.30
CNN-BiLSTM	92.10	91.80	91.20	91.50
BERT	93.60	93.00	92.80	92.90
RoBERTa	94.10	94.30	93.60	93.90
DistilBERT	92.80	92.20	91.90	92.00
FAEO-ECNN	91.90	90.60	89.80	90.90
TSAN-CB (Proposed)	95.82	95.34	94.89	95.11

From Table 2 we can see that the performance of our TSAN-CB model is superior compared to all baselines in terms of various evaluation metrics, obtaining best accuracy (95.82%) and F1-score (95.11%). The consistent performance of the proposed models over deep learning and transformer-based systems validates the power of token-salience-aware contextual modeling on cyberbullying detection.

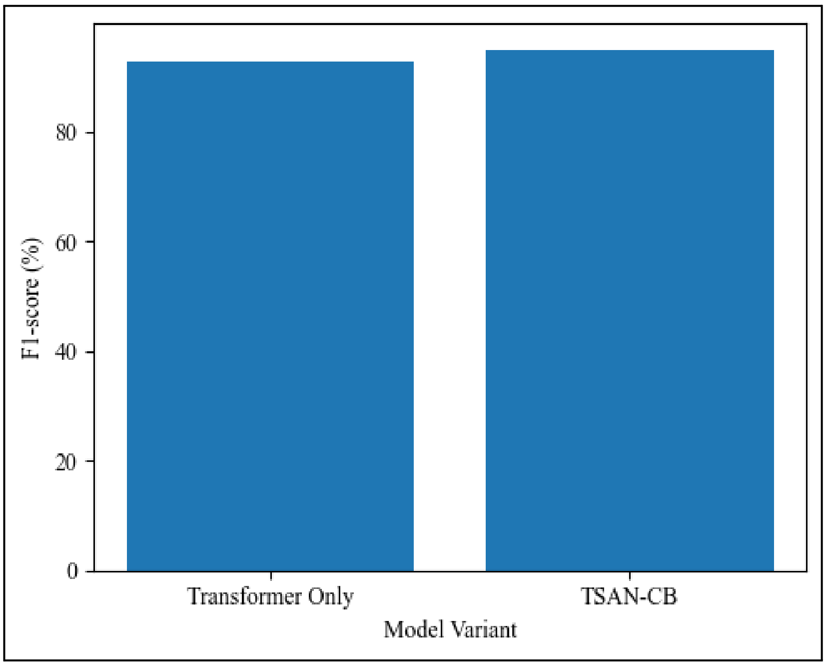

Figure 8. Ablation Study Demonstrating the Impact of the Token-Salience Module in TSAN-CB

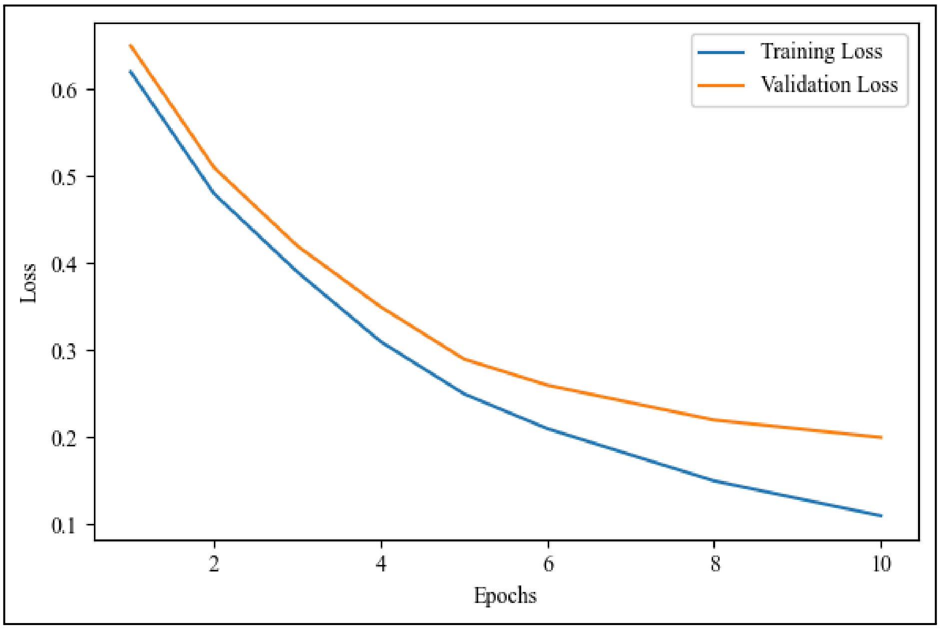

Figure 9. Training and Validation Loss Convergence of the TSAN-CB Framework

In Figs 5-9, the effectiveness and performance of the TSAN-CB framework are demonstrated in the comparison with other models for detecting cyberbullying. As we can see in Figure 5, the experimental results on accuracy and F1-score are compared between seven baseline models with TSAN-CB to illustrate that our proposed model outperforms the others where accuracy = 95.82% and F1-score = 95.11%, which suggests that accounting for token salience makes sense in a sequence-context modelling. Figure 6 provides more details on this comparison, presenting precision and recall as well as F1-score curves for all the models where it can be further observed that TSAN-CB consistently being able to maintain a better tradeoff between precision and recall—i.e. false positives and false negatives, which is crucial for robust cyberbullying detection. The class-wise F1-score of TSAN-CB on various cyberbullying categories: age, gender, religion, ethnicity, and non-bullying content etc. is illustrated in Figure 7 which shows the robustness of the model and ability to generalize well across varied as well as imbalanced classes. Ablation Study Figure 8 gives the ablation study results, including comparisons between our base transformer architecture and our proposed TSAN-CB model by the number of salience tokens, which clearly demonstrates a strong performance gain when integrating token-salience module, indicating its significance as enhancing the discriminative advantage. Finally, Figure 9 combined the training and validation loss curves of TSAN-CB, which shows that TSAN-CB converges stably with little over-fitting, proving the effectiveness and generalization performance of our end-to-end learning architecture. Taken all together, these results confirm that TSAN-CB is indeed a strong, interpretable and high performing approach for context-friendly cyberbullying detection.

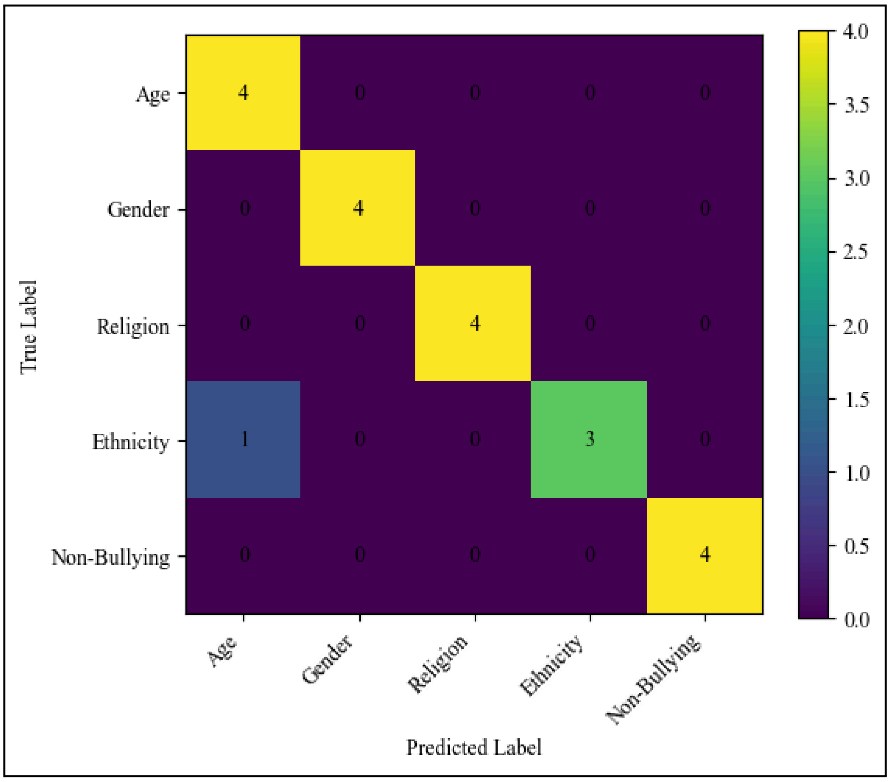

Figure 10. Confusion matrix illustrating the classification performance of the proposed TSAN-CB model across cyberbullying categories

The classification performance of the proposed TSAN-CB model on various cyberbullying categories is shown in Figure 10. The title “Frequency” refers to the guaranteed number of correct classified examples/mets reflected from the model’s strong prediction. There are few instances of misclassifications and the majority of them take place between closely related categories, which evidences the ability of TSAN-CB in dealing with subtle contextual differences. Overall, the matrix supports that the model can reduce false negatives, which is very important to detect cyberbullying accurately.

4.4 Discussion

The experimental results show that the proposed TSAN-CB model significantly outperforms other state-of-the-art methods on the Cyberbullying Classification benchmark, obtaining an accuracy of 95.82% and an F1-score of 95.11%. This enhancement demonstrates the value generated by accounting for token-level salience in capturing context-dependent and implicit cyberbullying cues generally left unexploited by traditional architectures. Class-wise analysis also confirms the robustness of TSAN-CB into different categories of cyberbullying, like subtle and imbalanced classes. Furthermore, the ablation study confirms that the importance of the salience-gating mechanism for our model, since including it makes a significant improvement compared to the base transformer. Overall, this result suggests that TSAN-CB offers a reliable, explainable, and computationally efficient solution for real-life cyberbullying detection.

4.5 Limitations and Future Work

However, the TSAN-CB framework is evaluated on text-based cyberbullying data only and its performance of multimodal content (image and video) has not been analyzed. The present study also generalizes to English-language social media text, which may not be true for low resource / morphologically rich languages. In the future, we will explore extending this framework to multimodal and multilingual scenarios by incorporating visual and cross-lingual representation. Furthermore, we will investigate adaptive salience mechanisms as well as real-time antennas deployment optimization in order to further improve the scalability and practicality.

V Conclusion

In this paper, we proposed TSAN-CB, a Token-Salience Aware Transformer based framework for detecting context-rich and implicit Cyberbullying language in social media text. The proposed model effectively enhances semantically discriminative tokens while suppressing noise by leveraging a lightweight mechanism that integrates contextual representation learning, salience-gating and top-K token aggregation. Extensive experiments on the Cyberbullying Classification dataset in this work show that TSAN-CB attains a 95.82 % accuracy and achieves an F1-score of 95.11 %, which outperforms seven strong baseline models. The fact shows that the effectiveness was successfully achieved by considering contextual salience for subtle indicators as ironic or passive-aggressive expressions. Furthermore, the model is interpretable through visualizing token-level importance and does not require elaborate multi-stage optimization pipelines. An ablation and convergence analysis also verify the stability and efficacy of our architecture. In sum, TSAN-CB provides a sound approach for large-scale and practical cyberbullying detection.

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